

SMRITI SATHI (स्मृति साथी)

Clinical Validity, Technical Rigor & Judge Defense Blueprint

Smart India Hackathon 2026 • Problem Statement SIH26003 • Ministry of Development of North Eastern Region (MDoNER)

SIH 2026 IDEA DEFENSE

TEAM CTRALTELITE

OFFICIAL PREPARATION
GUIDE

⚠ The Single Most Dangerous Question Judges Will Ask:

"You are collecting cognitive performance data and generating alerts. On what basis can you say this indicates cognitive decline, and how do you prevent the system from falsely labeling someone as declining?"

🛡 The Master Defense:

"Smriti Sathi is strictly a **supportive, non-pharmacological cognitive engagement and longitudinal monitoring tool** grounded in Cognitive Stimulation Therapy (CST) and the DAHLIA study—it **does NOT** diagnose dementia.

To prevent false decline alarms, we utilize a **14-session moving median baseline**. An alert is NEVER triggered by a bad day or temporary illness; it requires a statistically significant drop (>2 standard deviations across ≥7 consecutive sessions) combined with repeated missed routine reminders. Crucially, the system does not diagnose—it generates an **Actionable ASHA Check-in Signal**, prompting a human healthcare worker to perform a standard clinical screening (HMSE/MoCA) or connect with a Primary Health Centre (PHC) medical officer."

1. Cognitive Assessment Methodology: What We ACTUALLY Measure

Unlike simple entertainment games, Smriti Sathi measures 5 specific, clinically grounded cognitive markers per workout:

Cognitive Domain	Game / Module	Exact Parameter Measured	Clinical Correlation
Working Memory	Speed Match (Card Pairing)	Recall Accuracy (% correct on 1st vs 2nd look) & Attempt Multiplier	Visual retention & hippocampal spatial recall stability
Processing Speed	Speed Match / Math Workout	Motor-Visual Latency (milliseconds from render to touch)	Psychomotor processing speed & gentle reflex tracking
Temporal Orientation	Masterpiece (Daily Routine)	Chronological Sequencing Errors (Morning → Night sequence)	Executive functioning, time disorientation, daily confusion
Attention & Focus	Language & Proverbs Recall	Semantic Precision & Distractor Rejection Rate	Semantic memory preservation & cultural reminiscence anchor
Care Adherence	Care Reminders	Prompt Acknowledgment Latency & Consistency (% completed)	Medication consistency, hydration, functional daily living (ADL)

2. Demystifying the "AI" Claim: Where AI Lives vs. Why the Core is Rule-Based

Where AI Actually Lives Today (NLP & Speech):

We leverage the Government of India's **Bhashini AI Platform** (MeitY) for Automatic Speech Recognition (ASR), Text-to-Speech (TTS), and Neural Machine Translation (NMT) in low-resource North

Eastern languages (Assamese, Bodo, Manipuri). This enables true voice-first interaction for illiterate and semi-literate rural elders.

Why the On-Device Cognitive Engine is Intentionally Rule-Based (The 3-Up / 2-Down Staircase):

Judges may ask: "*Why isn't the cognitive difficulty engine deep learning?*" The defense is intentionality:

1. **Explainability & Clinical Safety:** Elder health algorithms must be deterministic and transparent. A deep learning black box cannot be clinically audited by an ASHA worker or doctor.
2. **Extreme Edge Compute Constraints:** The app must run offline on ₹5,000 Android devices with 1–2GB RAM. A rule-based staircase engine operates at 0% CPU overhead, zero thermal throttling, and zero battery drain.
3. **Future AI Roadmap (Phase 2):** An on-device quantized TFLite Anomaly Detection model (Isolation Forest / Autoencoder) trained on touch dispersion jitter, circadian play regularity, and multi-week trend decay.

3. The Complete ASHA Worker Field Workflow

The solution is not merely a patient app; it is a closed-loop public health pipeline bridging rural homes to the state healthcare system:

Step 1

Elderly Daily Practice (100% Offline): The elder plays daily 5-minute CST workouts on a bedside tablet. All scores, response times, and reminder confirmations are saved locally to encrypted IndexedDB.

Step 2

ASHA Worker Routine Visit: The ASHA worker visits the home during routine weekly rounds with their standard mobile smartphone.

Step 3

Zero-Internet P2P Bluetooth Sync: The tablet pairs with the ASHA worker's phone via Bluetooth Low Energy (BLE GATT service 0xFE26). 14 days of telemetry (approx. 25 KB) synchronizes in <3 seconds without internet.

Step 4

Caregiver Dashboard Assessment: The ASHA worker inspects the Smriti Performance Index (SPI), 10-session trend curve, and medication adherence percentage.

Step 5

Signal Evaluation & Family Counseling: If a persistent multi-session decline signal is present, the ASHA worker discusses recent changes (sleep, nutrition, disorientation) with family caregivers.

Step 6

PHC Escalation: When an alert is verified, the ASHA worker logs the case for the Primary Health Centre (PHC) Medical Officer for formal cognitive screening.

4. Cultural Localization & Community Validation Methodology

How we validate that regional cultural anchors actually work for North Eastern dementia patients:

Validation Pillar	Methodology & Evidence	Implementation in Smriti Sathi
Linguistic Fidelity	Native dialectal validation by regional linguists & ASHA workers	Curated vocabulary in Assamese (অসমীয়া), Bodo (বৰ), and Manipuri (মৈতৈলোন); regional pronunciation phonetics.
Reminiscence Anchors	Grounded in Reminiscence Therapy (RT) principles (long-term memory preservation)	Using regional folk symbols (Assamese Dhol, Bihu motifs, Losar patterns, local bazaar items) rather than alien Western symbols.
Errorless Scaffolding	Eliminates cognitive frustration and behavioral agitation in dementia	Pulsing gold hints on Level 1; zero buzzers, zero failure screens; spoken gentle encouragement.

5. Privacy Architecture (DPDP Act 2023 & ABDM Alignment)

Patient / Guardian Consent → Local AES Encryption → Ephemeral Audio → Audit & Revocation

- **What is Stored:** Anonymized numerical session metrics (timestamps, accuracy, response latencies, reminder checkmarks).
- **What is NEVER Stored:** Raw audio recordings are processed ephemerally in RAM and never saved to disk or transmitted.
- **Lost Device Safeguard:** On-device database is encrypted at rest; role-based access separates elder game views from caregiver metrics.
- **DPDP Revocation:** One-tap "Revoke Consent & Erase Data" button in Settings immediately purges all local records.

6. Concrete Evaluation KPIs (Measurable Impact)

Cognitive Engagement

Baseline: 0 sessions/wk
Target: ≥5 sessions/wk
Measured via local session logs

Medication Adherence

Baseline: ~40% in rural NER
Target: >80% adherence
Measured via voice reminder acks

ASHA Review Time

Baseline: 30 mins manual query
Target: <5 mins via BLE sync
Measured during pilot visits

7. Competitor & Category Differentiation Matrix

Core Dimension	Western Brain Training (Lumosity, Elevate)	Clinical Apps (CogniFit)	Govt Tele-MANAS	SMRITI SATHI (SIH26003)
Target Demographic	Healthy, tech-savvy urban adults	Urban patients with clinical supervision	General distress calling	Rural NER elderly with MCI & Dementia
Offline Capability	Requires continuous internet	Requires continuous internet	Requires active telephone line	100% Offline-First (zero internet for games)
Language Support	English, Spanish, major languages	English, Western languages	Major Indian official languages	Assamese, Bodo, Manipuri & NER dialects (Bhashini AI)
Cost / Barrier	\$12 – \$15/month subscription	\$20+ /month subscription	Free helpline	100% Free & Open Source (Zero Paywalls)
Community Workflow	Solo user only	Specialist clinic only	Centralized call center	Integrated ASHA Worker BLE Sync & Family Dashboard
Accessibility	Small text, fast timers, punitive	Standard clinical UI	Voice only	56–88px targets, high contrast, errorless learning

8. Clinical Evidence Mapped Directly to UI/UX Decisions

Published Evidence & Study	Key Research Finding	Concrete Smriti Sathi Architecture Decision
Aguirre et al. (Cochrane 2013)	Cognitive Stimulation Therapy (CST) consistently improves cognitive scores & quality of life in mild-to-moderate dementia.	Structured 4-domain daily workout cycle (Memory, Orientation, Math, Language) limited to 5–10 mins to avoid fatigue.
DAHLIA Study (Rural India)	High digital illiteracy among rural Indian elderly; text-heavy apps fail; voice and visual guidance drive adoption.	Voice-first prompts in native dialects; visual icon badges; minimum 56px touch buttons; no keyboard typing needed.
Observer Research Foundation (NER Connectivity)	Over 60% of interior North Eastern villages experience persistent cellular blackouts and intermittent power.	Local-first IndexedDB storage; opportunistic BLE sync; zero cloud dependency for patient-facing workouts.

9. Deployment Economics (Why It Is Feasible at Scale)

- **Hardware Independence:** Runs on existing family Android phones or low-cost ₹5,000 commodity tablets (Android 7.0+).
- **Zero Data Cost for Families:** Because workouts operate 100% offline, impoverished rural families do not require active mobile data plans.
- **Scalable Cloud Economics:** A single \$7/month cloud companion instance handles intermittent synchronization for 10,000+ registered elders.

- **Public Health Integration:** Distributable as a standalone sideloaded APK via Primary Health Centres (PHC) and District Health Societies.

10. Staged Validation & Delivery Roadmap

Phase 1: Working Functional Prototype (Completed & Demonstrated)

Native Android APK (4.3 MB), 4 playable games, on-device adaptive staircase, care reminders, BLE sync simulation, caregiver dashboard.

Phase 2: Community Field Pilot (Months 1–3)

Deployment across 50 rural households in Kamrup district, Assam, in collaboration with 10 ASHA workers; measuring task completion & adherence.

Phase 3: Supervised Clinical Correlation Study (Months 4–6)

Collaboration with Department of Neurology at Guwahati Medical College & Hospital (GMCH) to benchmark SPI trends against formal HMSE/MoCA scores.

Phase 4: Health System Integration (Months 7–12)

Ayushman Bharat Digital Mission (ABDM) Health ID integration and regional rollout across Assam, Meghalaya, and Manipur.